

Development of reproductive system

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Introduction

- The reproductive system is a collection of **internal and external sex organs** which work together for the purpose of **sexual reproduction**
- The **development** of these reproductive organs **begins at an early stage in the embryo**
- The **genotype** of the embryo (46,XX or 46,XY) **is established at fertilization**
- **During weeks 1–6**, the embryo remains in a **sexually indifferent** or undifferentiated stage → **genetically female embryos and genetically male embryos are phenotypically indistinguishable**

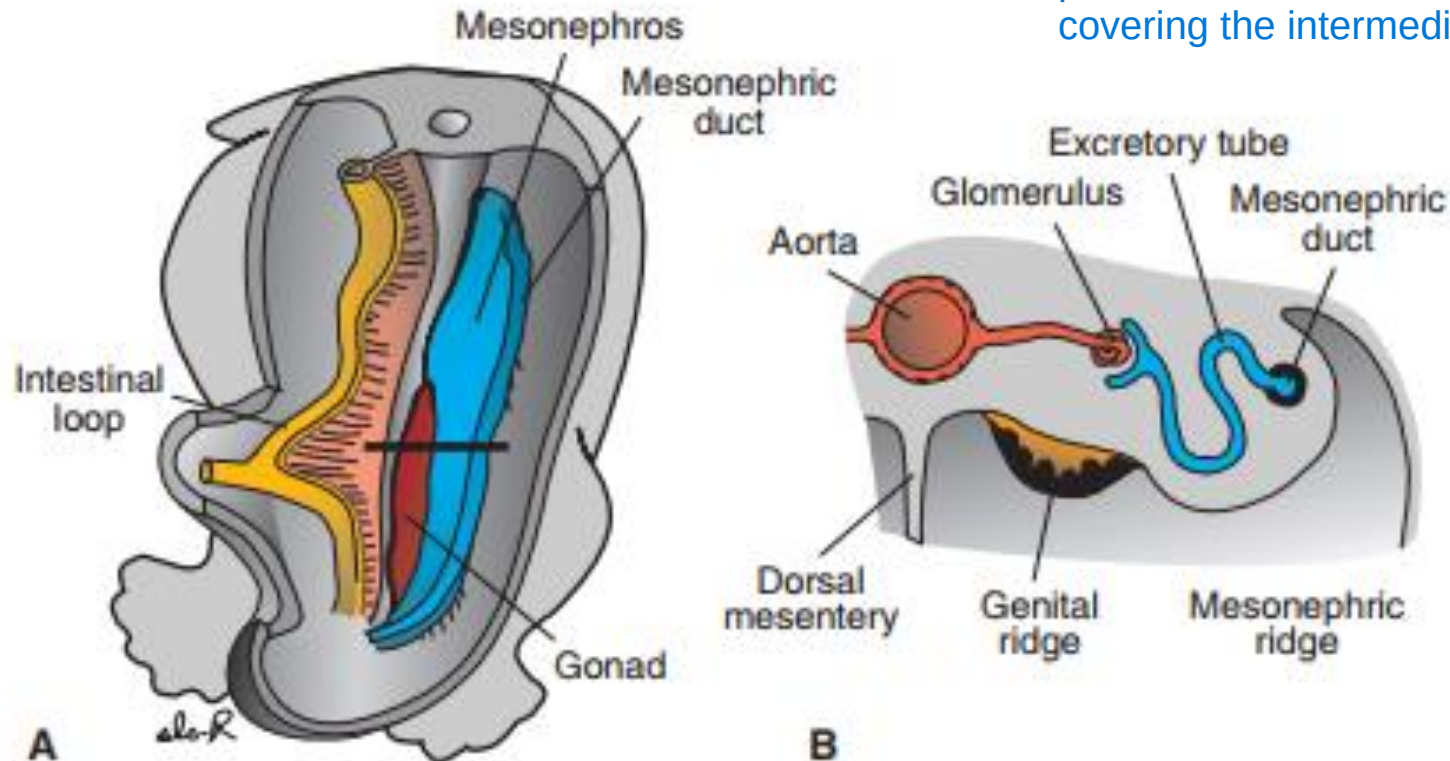
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- During **week 7** → the indifferent embryo begins phenotypic sexual differentiation.
- By **week 12** → female or male characteristics of the external genitalia can be recognized beginning of second trimester
- By **week 20** → phenotypic differentiation is complete
- Sexual differentiation is a **complex process** that involves many genes including some that are **autosomal** (FGF9, Sox9, WT1, SF1, Lim1, WNT4...) fibroblast growth factor9
- The **key to sexual dimorphism** is the **Y chromosome** which contains the **testis determining factors (TDF)** called **SRY** on its short arm (Yp11)
- There is a close link throughout with the **development of the urinary system**

Gonads: Indifferent stage

- In the first stage of gonadal development, it is impossible to distinguish between the male and female gonad
- The gonads begin as **genital ridges** – a pair of longitudinal ridges derived from **intermediate mesoderm** and **overlying epithelium**
 - They **initially do not contain any germ cells**

The urogenital ridge arises as a dorsal body wall thickening that harbors populations from both the primitive mesoderm and the coelomic epithelium covering the intermediate mesoderm



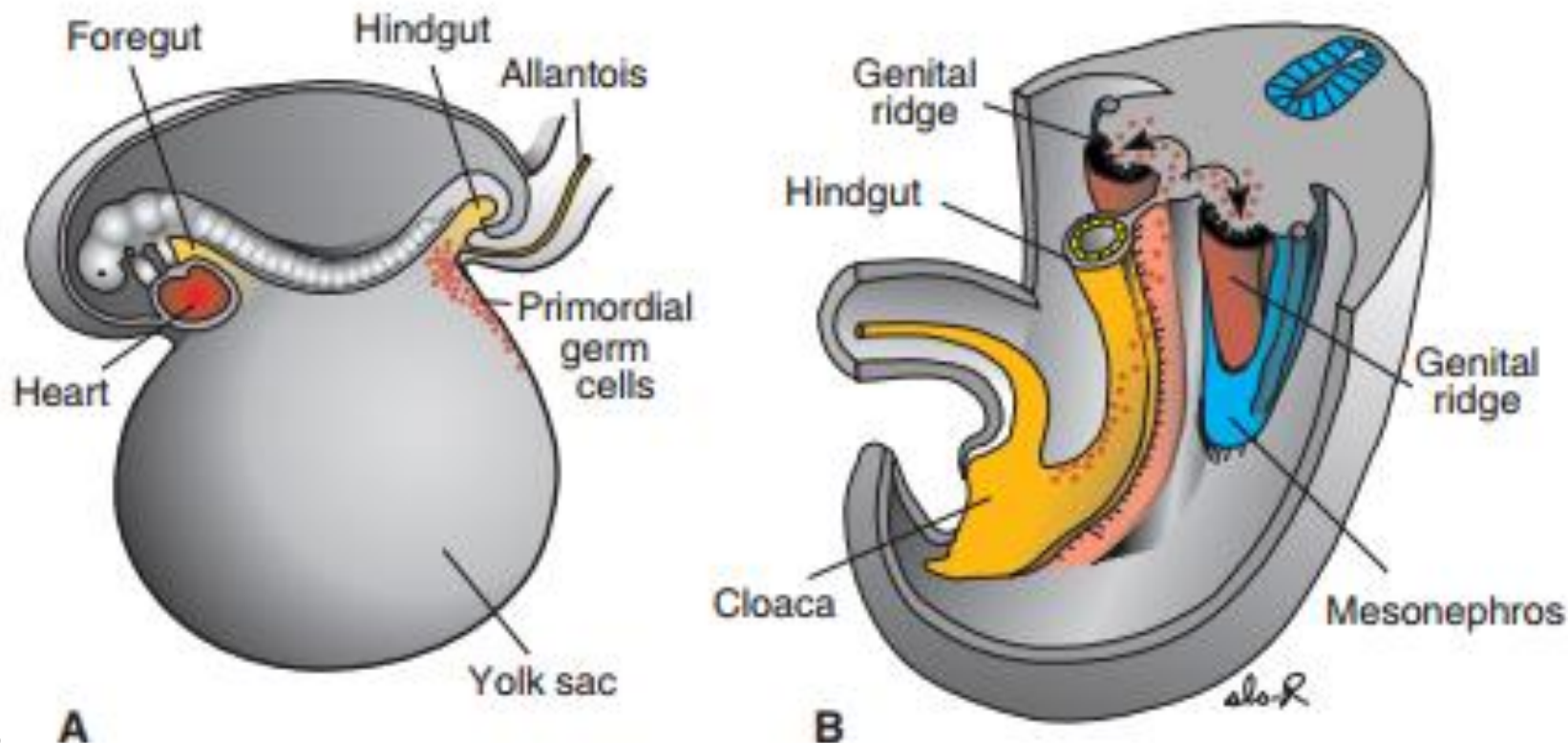
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primordial germ cells are derived from pluripotent epiblast cells in the extra-embryonic structure during week 2

- In the **fourth week**, germ cells begin to migrate from the endoderm lining of the **yolk sac** to the genital ridges, via the **dorsal mesentery of the hindgut**

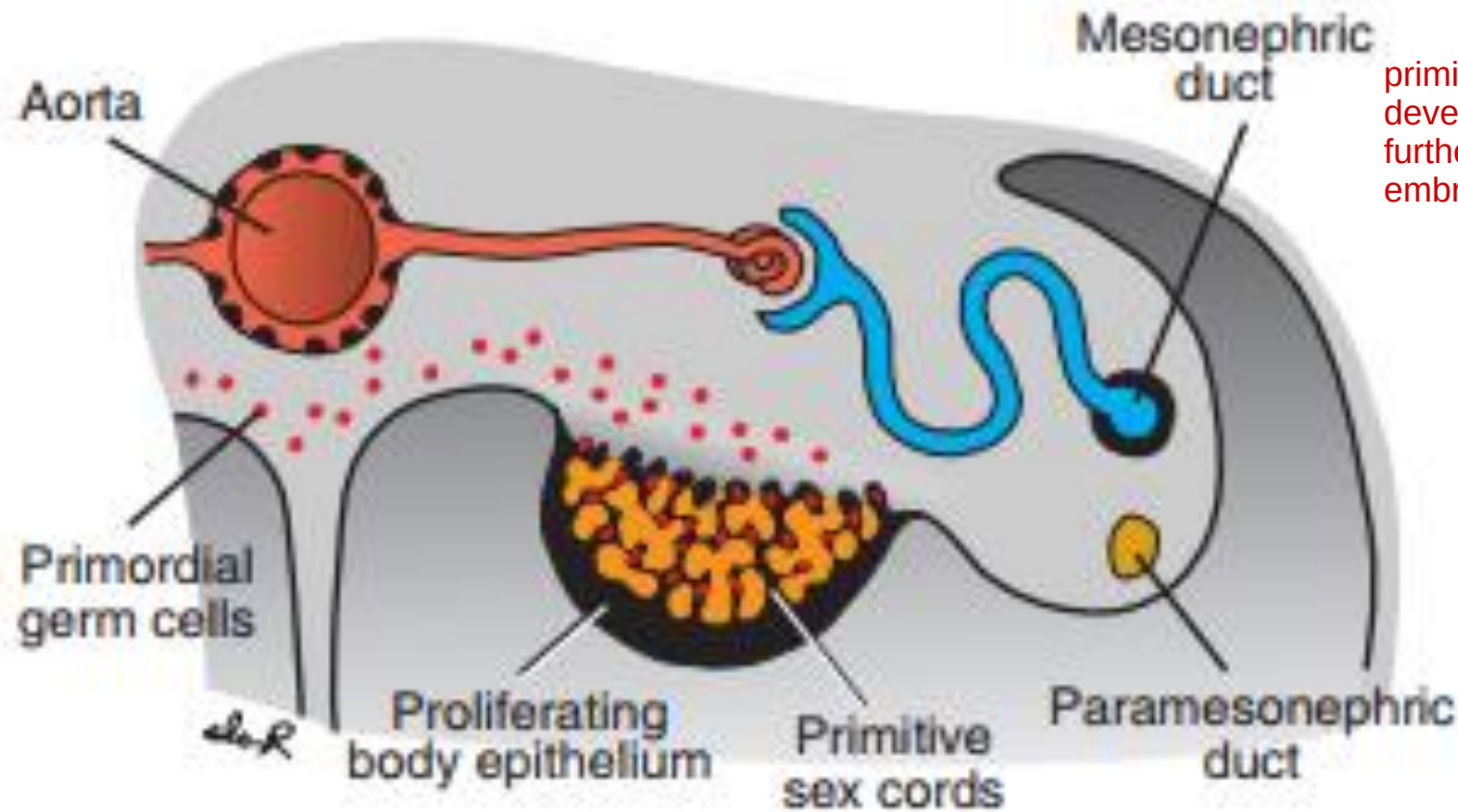
➤ They reach the genital ridges in the **sixth week**

during their migration, they undergo a series of mitotic cell division to increase their number



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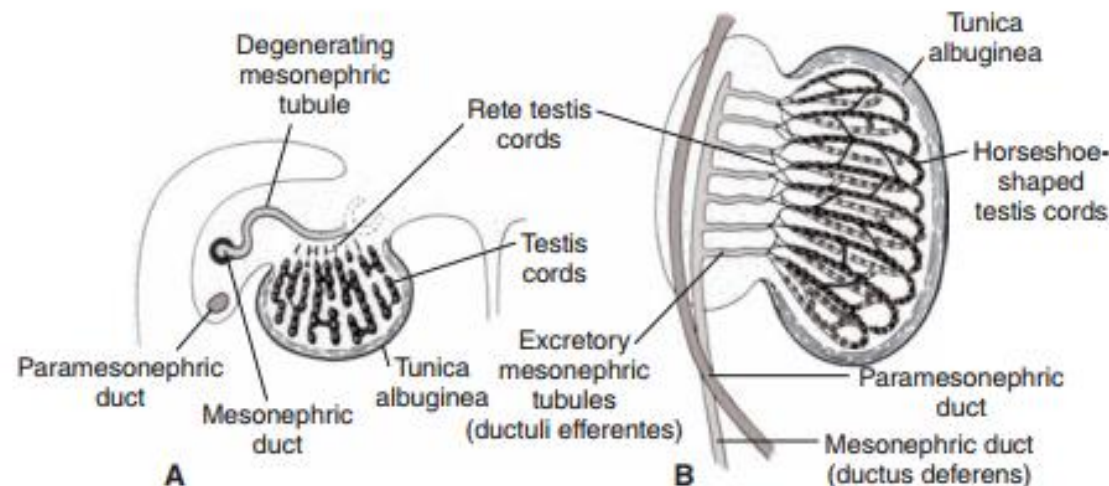
- Simultaneously, the epithelium of the genital ridges proliferates and penetrates the intermediate mesoderm to form the **primitive sex cords**
- The combination of **germ cells** and **primitive sex cords** forms the **indifferent gonad** – from which development into the testes or ovaries can occur



primitive sex cords are structures that develop from the genital ridges that further differentiate based on an embryo's sex.

Testes

- In a male embryo, the **XY sex chromosomes** are present
- The Y chromosome contains the **SRY gene**, which stimulates the development of the **medullary cords** to form **testis**
- The **tunica albuginea**, a fibrous connective tissue layer, forms around the cords
- A portion of the testis cords breaks off to form the future **rete testis** intratesticular duct
- The remaining cords contain two types of cells:
 - **Germ cells**
 - **Sertoli cells** (derived from the surface epithelium of the gland)



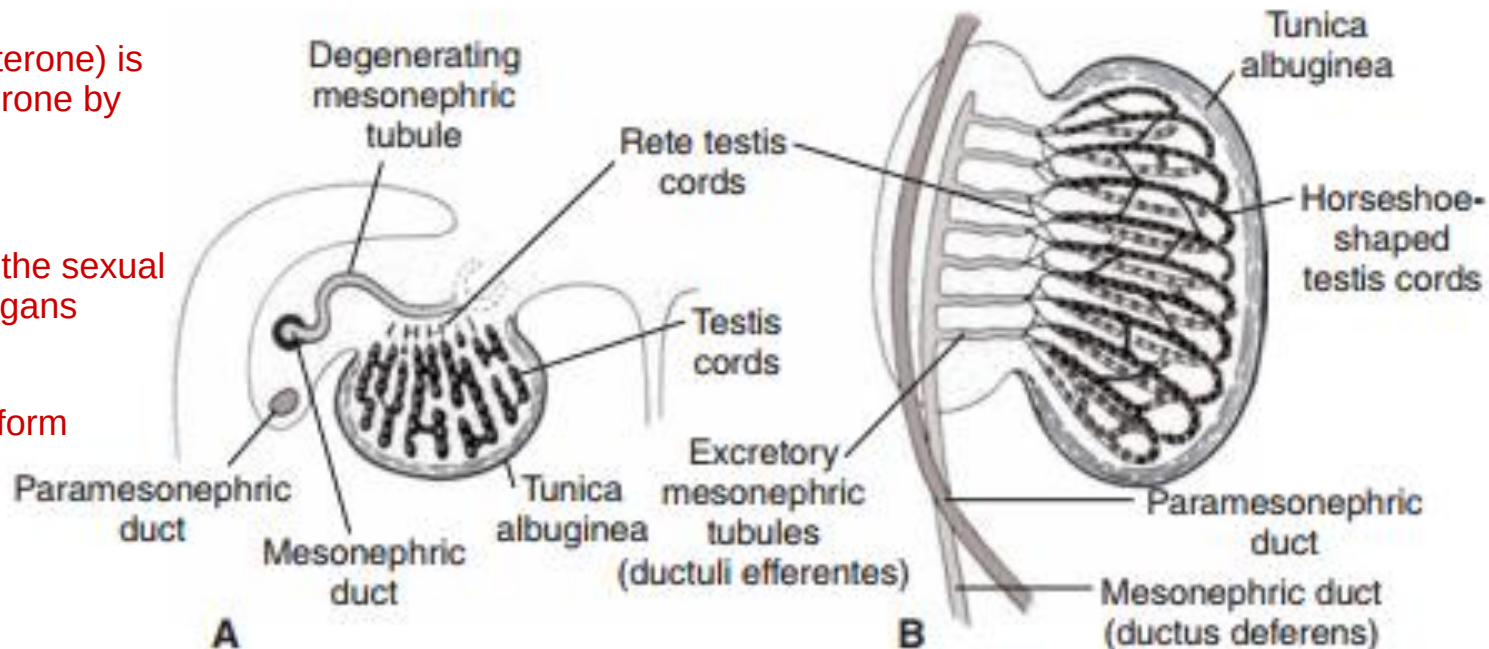
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- At puberty, these ^{canalization due to apoptosis} cords acquire a lumen and become the **seminiferous tubules** – the site within which sperm will be formed
- Located between the testis cords are the **Leydig cells** (derived from the intermediate mesoderm)
- In the eighth week, they begin production of **testosterone** – which drives differentiation of the internal and external genitalia

DHT(dihydrotestosterone) is made from testosterone by 5-alpha-reductase

DHT is involved in the sexual differentiation of organs

DHT is the active form

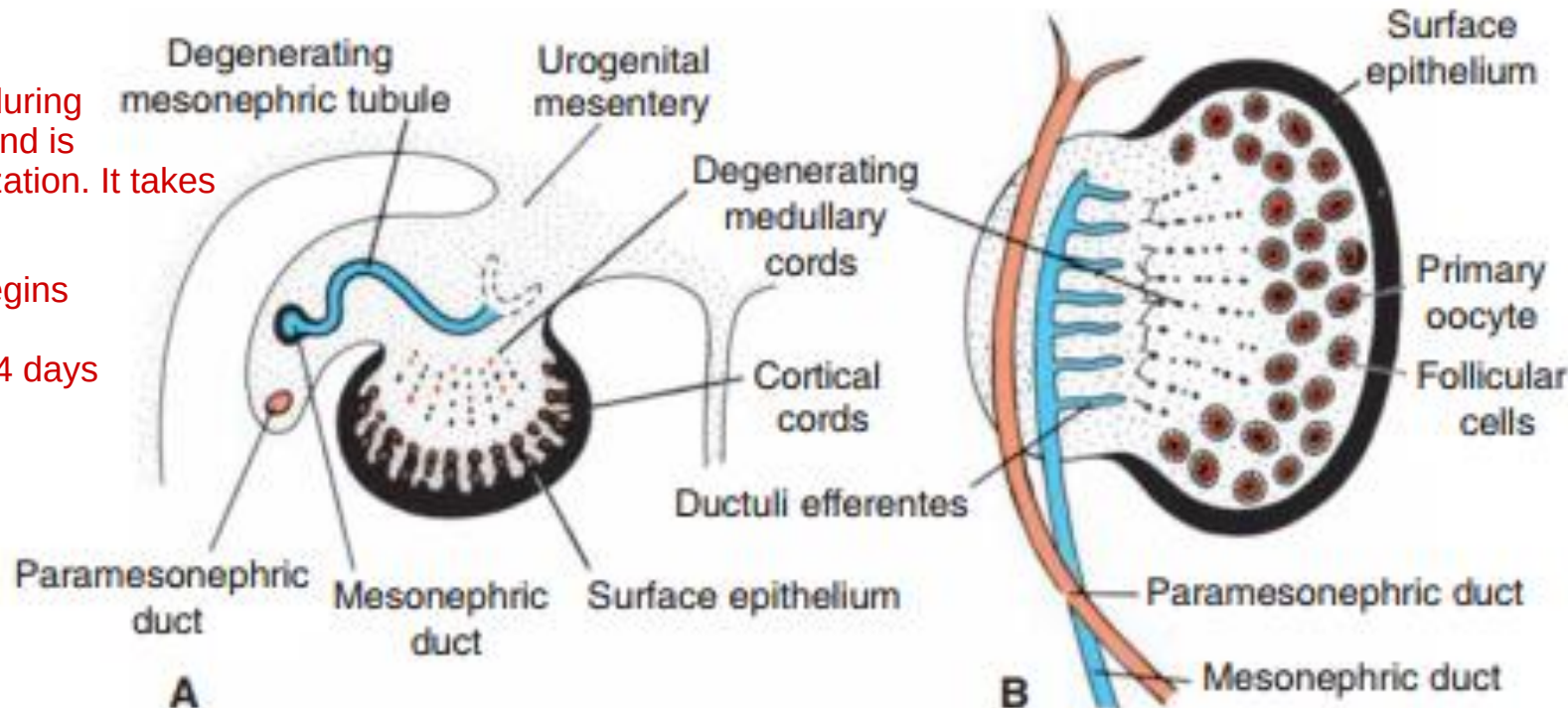


Ovaries

- In a female embryo, the **XX** sex chromosomes are present
- As there is **no Y chromosome**, there is **no SRY gene** to influence development
- Without it, the medullary cords **degenerate** and do not form the testes
- Instead, the **epithelium** of the gonad continues to proliferate, producing **cortical cords**
- In the **third month**, these cords break up into clusters, surrounding each **oogonium** (germ cell) with a **layer of epithelial follicular cells**, forming a **primordial follicle**

oogenesis begins during embryonic period and is completed at fertilization. It takes several years

spermatogenesis begins after puberty and it takes about 74 days (2 months and half)

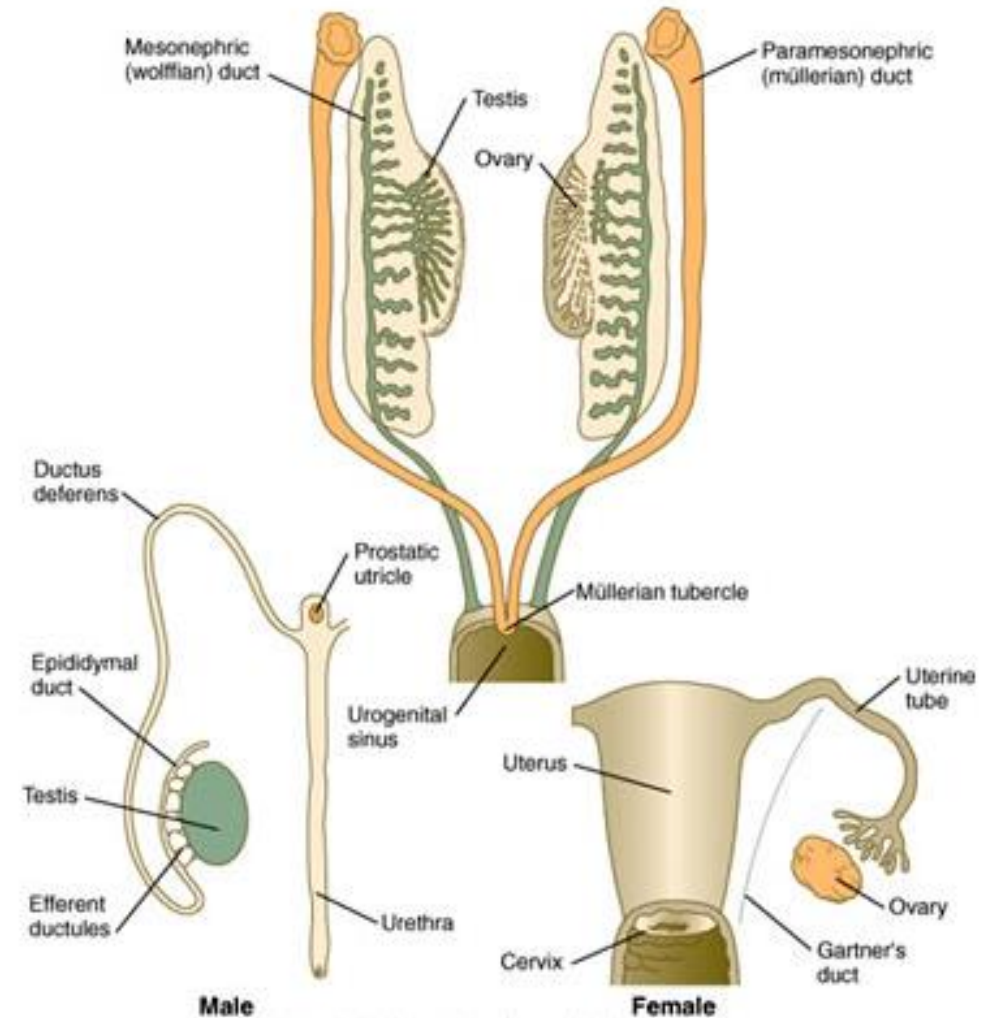


first meiosis of oogonia is arrested at prophase 1 until puberty

no oogonia is present at birth. they are all changed into primary oocytes

Internal genitalia: Indifferent stage

- In the early weeks of urogenital development, all embryos have two pairs of ducts, both ending at the cloaca. These are the:
 - **Mesonephric (Wolffian) ducts** found medially
 - **Paramesonephric (Müllerian) ducts** found laterally

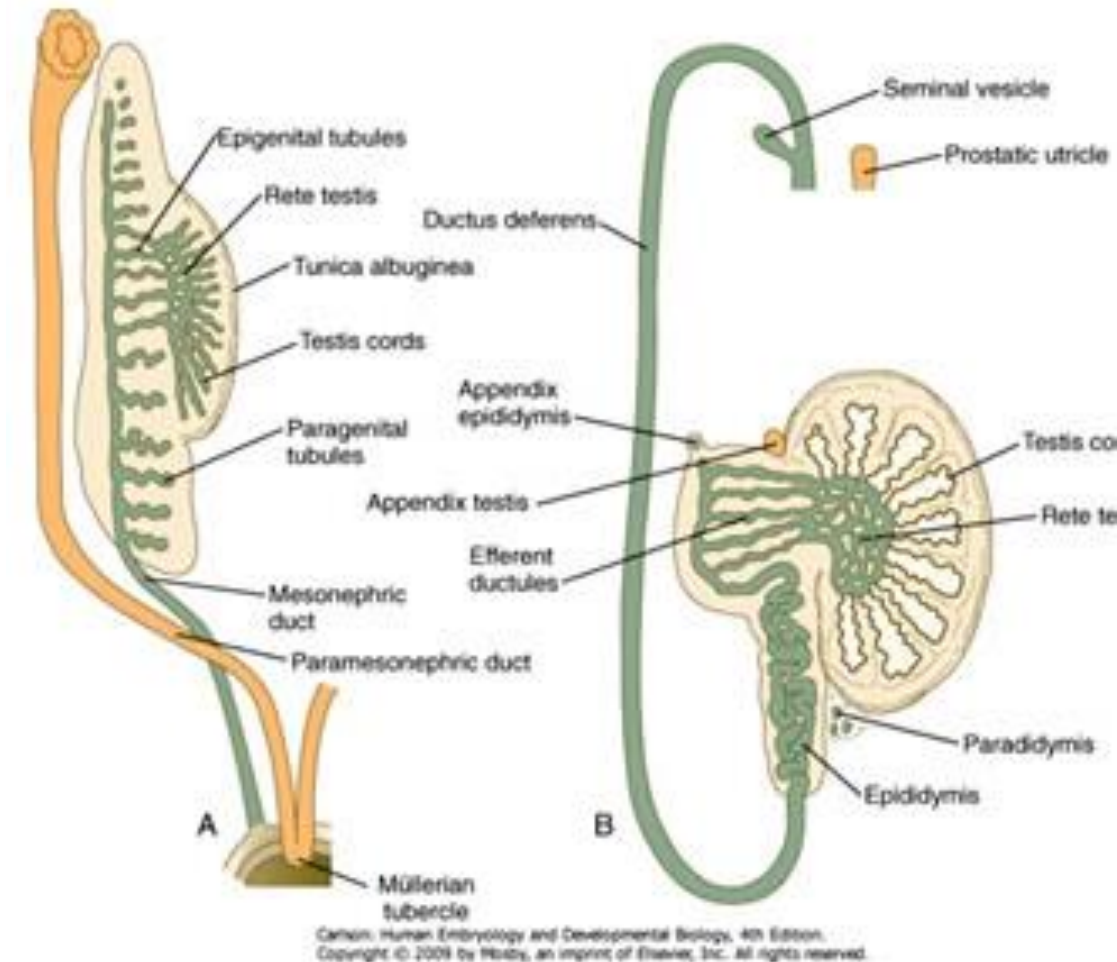


Male

epididymis= site of storage and maturation of sperm cells
seminal vesicles= secrete most of the semen content

- In the presence of **testosterone** (produced by the Leydig cells), the **mesonephric ducts** develop to form the primary male genital ducts
 - They give rise to the **efferent ductules**, **epididymis**, **vas deferens** and **seminal vesicles**
- Meanwhile, the **paramesonephric ducts** degenerate in the presence of **anti-Müllerian hormone** – produced by **Sertoli cells** in the testes
 - Its developmental remnant is the **appendix testis**; a small portion of tissue located on the upper pole of each testicle, which has no physiological function

blood testes barrier??



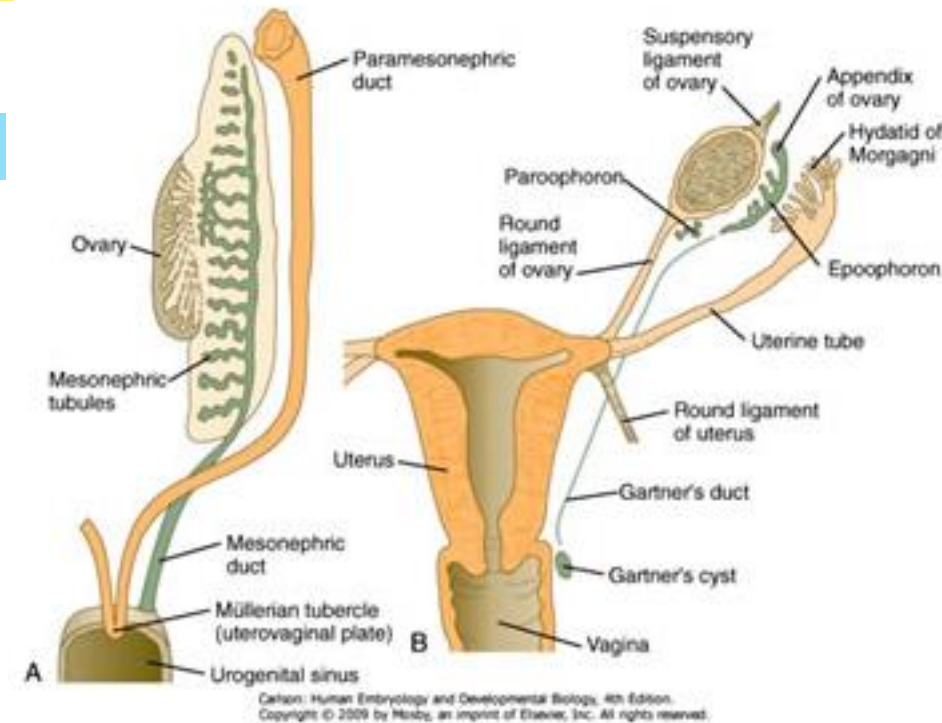
Female

- In the female, there are **no Leydig cells to produce testosterone** → **the mesonephric ducts degenerate**, leaving behind only a vestigial remnant – **Gartner's duct**
- Equally, the **absence of anti-Müllerian hormone** also allows for development of the **paramesonephric ducts**
- Initially, these ducts can be described as having three parts:
- **Cranial** – becomes the **Fallopian tubes**
- **Caudal** – fuses to form the uterus, cervix and upper 1/3 of the vagina
- The **lower 2/3 of the vagina** is formed by **sinovaginal bulbs** (derived from the **pelvic part of the urogenital sinus**)

mesodermal in origin

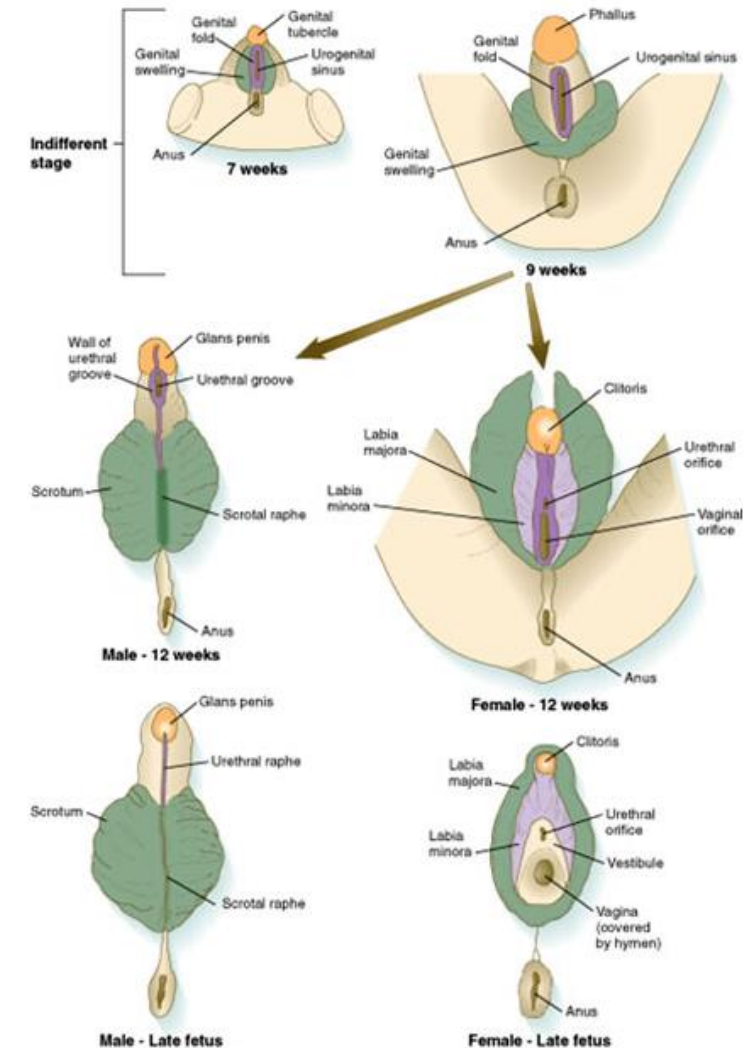
endodermal in origin

the muscular and connective tissue wall is derived from the mesoderm
only the epithelial lining is from endoderm



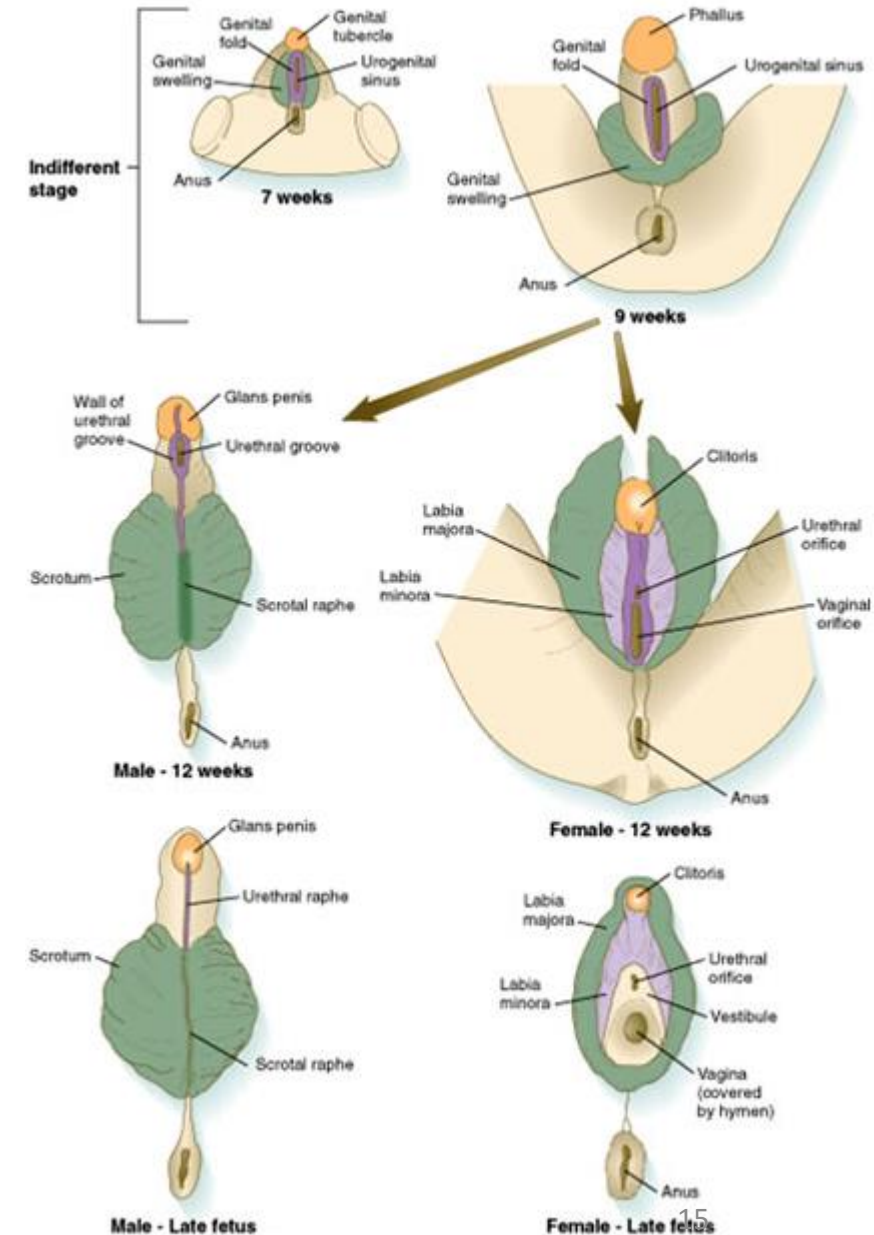
External Genitalia: Indifferent stage

- The development of the external genitalia begins in the third week
- Mesenchymal cells from the primitive streak migrate to the cloacal membrane to form a pair of **cloacal folds**
- Cranially, these folds fuse to form the **genital tubercle**
- Caudally, they divide into the urethral folds (anterior) and anal folds (posterior)
- **Genital swellings** develop either side of the urethral folds



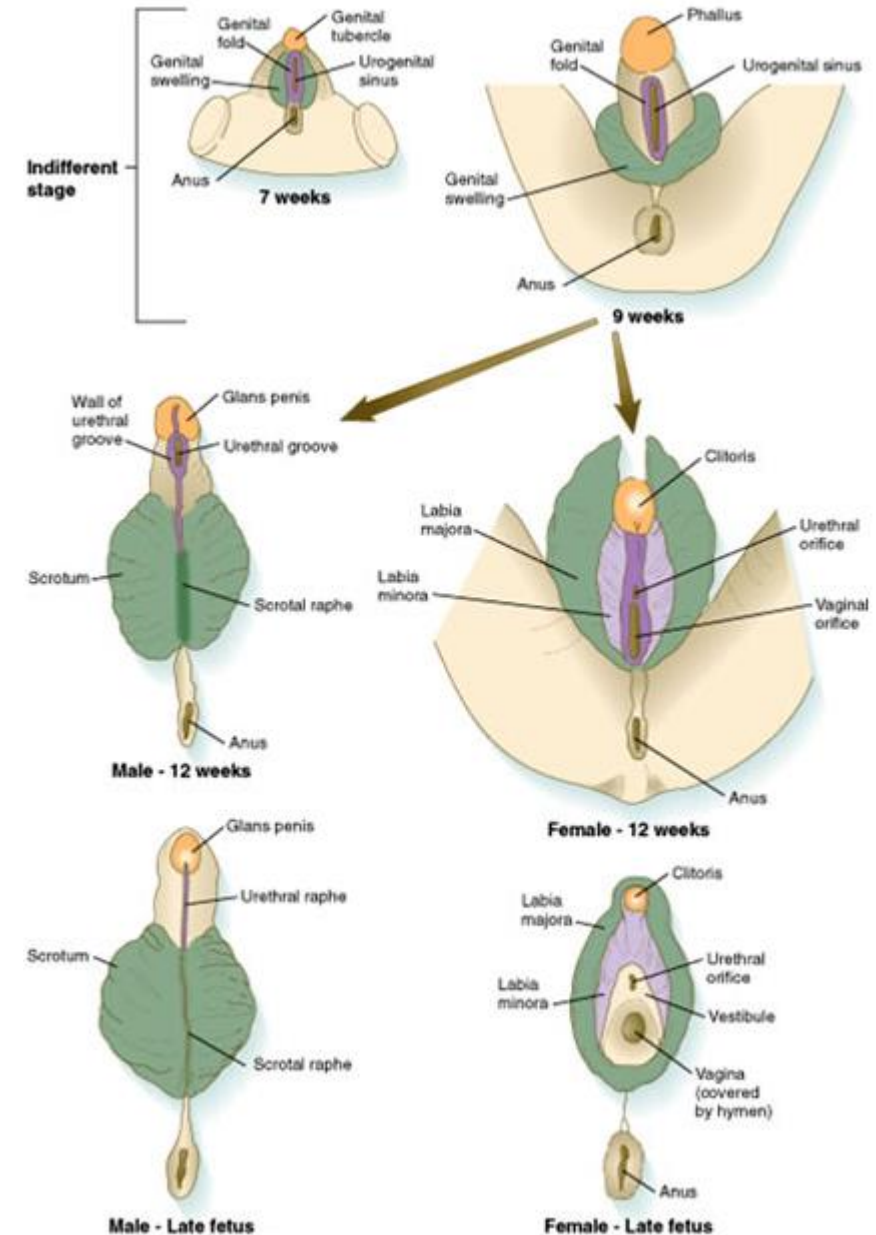
Male

- The development of the indifferent genitalia into the male genitalia is driven by the presence of androgens from the testes, namely **dihydrotestosterone (DHT)**
- There is rapid elongation of the genital tubercle, which becomes the **phallus**
- The **urethral folds** are pulled to form the **urethral groove** – this extends along the caudal aspect of the phallus
- The folds close over by the 4th month, forming the **penile urethra**
- The **genital swellings** become the **scrotal swellings**, moving caudally to eventually form the **scrotum**



Female

- Estrogens in the female embryo are responsible for external genital development
- The genital tubercle only elongates slightly to form the clitoris
- The urethral folds and genital swellings do not fuse, but instead form the labia minora and labia majora respectively
- The urogenital groove therefore remains open, forming the vestibule into which the urethra and vagina open



Descent of gonads

- While the gonads arise in the **upper lumbar region**, they are each tethered to the **scrotum** or **labia** by the **gubernaculum** – a ligamentous structure formed from mesenchyme

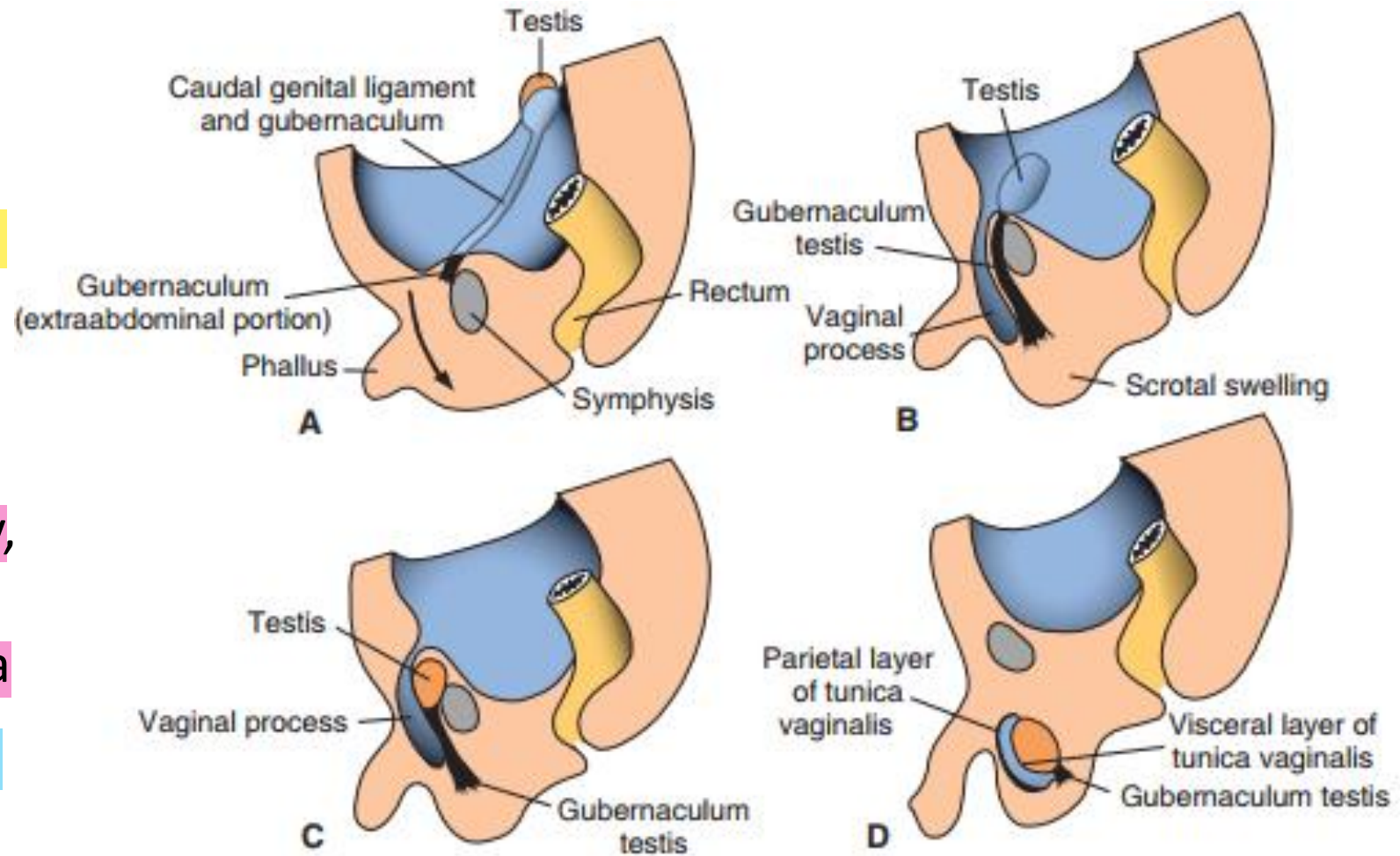
The outer layer of peritoneum, the processus vaginalis (tunica vaginalis reflexa), is reflected on to the testis to form the serous outer layer of that organ, the tunica vaginalis propria.

The processus vaginalis is a blindended evagination of the abdominal wall that develops during fetal life and typically undergoes obliteration in early life.

Descent of testes

is evident 3 months after birth

- As the body of the fetus grows, the testes become **more caudal**
- They pass through the inguinal canal around the 28th week, and reach the scrotum by the 33rd week
- During their descent, the testes retain their original blood supply, with the testicular arteries branching from the lumbar aorta
- The scrotal ligament is the adult remnant of the gubernaculum



the scrotal ligament secures the testis to the most inferior portion of the scrotum, tethering it in place and limiting the degree to which the testis can move within the scrotum. Abnormal function of the scrotal ligament can allow for testicular torsion to occur.

Descent of ovaries

- The ovaries initially migrate caudally in a similar fashion to the testes from their origin on the posterior abdominal wall
- However they do not travel as far, reaching their final position just within the true pelvis
- The gubernaculum becomes the ovarian ligament and round ligament of the uterus

the ovarian ligament is between the cornu of the uterus and the ovary

the round ligament of the uterus is between the cornu of the uterus and the labia majora

uterine horns= cornu of the uterus

Malformations of the genital system: ASD

❑ Turner's Syndrome (Gonadal Dysgenesis)

- Turner's syndrome results from a chromosomal anomaly (45,XO)
- Individuals with this syndrome possess primordial germ cells that degenerate shortly after reaching the gonads
- In the absence of gonadal hormones, the genitalia develop along female lines, but remain infantile

True Hermaphroditism

- is an extremely rare condition, possess both testicular and ovarian tissue
- Most true hermaphrodites have a 46,XX chromosome constitution, and the external genitalia are basically female, although typically the clitoris is hypertrophied

Female Pseudohermaphroditism

- The internal genitalia are typically female, but the external genitalia are masculinized, either from excessive production of androgenic hormones by the adrenal cortex (congenital virilizing adrenal hyperplasia)
- The degree of the external masculinization can vary from simple clitoral enlargement to partial fusion of the labia majora into a scrotumlike structure

Male Pseudohermaphroditism (46, XY)

- Results from inadequate hormone production by the fetal testes, the phenotype can vary
- It is often associated with small penis hypoplasia of the phallus, and there may be various degrees of persistence of paramesonephric duct structures

Testicular Feminization (Androgen Insensitivity) Syndrome

- Are genetic males (46,XY) and possess internal testes, but they typically have a normal female external phenotype and are raised as females
- Often, testicular feminization is not discovered until the individual seeks treatment for amenorrhea or is tested for sex chromatin before athletic events

Androgen insensitivity (testicular feminization) syndrome is a rare inherited form of male pseudohermaphroditism that occurs in phenotypically normal women

DSD - 5 α -reductase deficiency (46, XY)

- Occurs in genetic males with mutations in 5 α -reductase necessary for the conversion of testosterone to the more potent androgen, dihydrotestosterone (DHT)
- External genitalia are partially affected
- Mesonephric ducts are intact since they only require testosterone (which is still present)
- Normal production of MIS from Sertoli cells causes Paramesonephric duct regression

DSD - Persistent Müllerian Duct syndrome (46, XY)

- Occurs in genetic males with mutations in MIS or the MIS receptor
- Because of testosterone and DHT production, there are normal male external genitalia and male genital ducts
- Because there is effectively NO Mullerian inhibition, the paramesonephric ducts PERSIST; i.e. there is a small uterus and paired fallopian tubes
- The testes may lay either in what would be the normal position for ovaries (i.e. within the broad ligament) or one or both testes may descend into the scrotum

Abnormalities of uterus and vagina

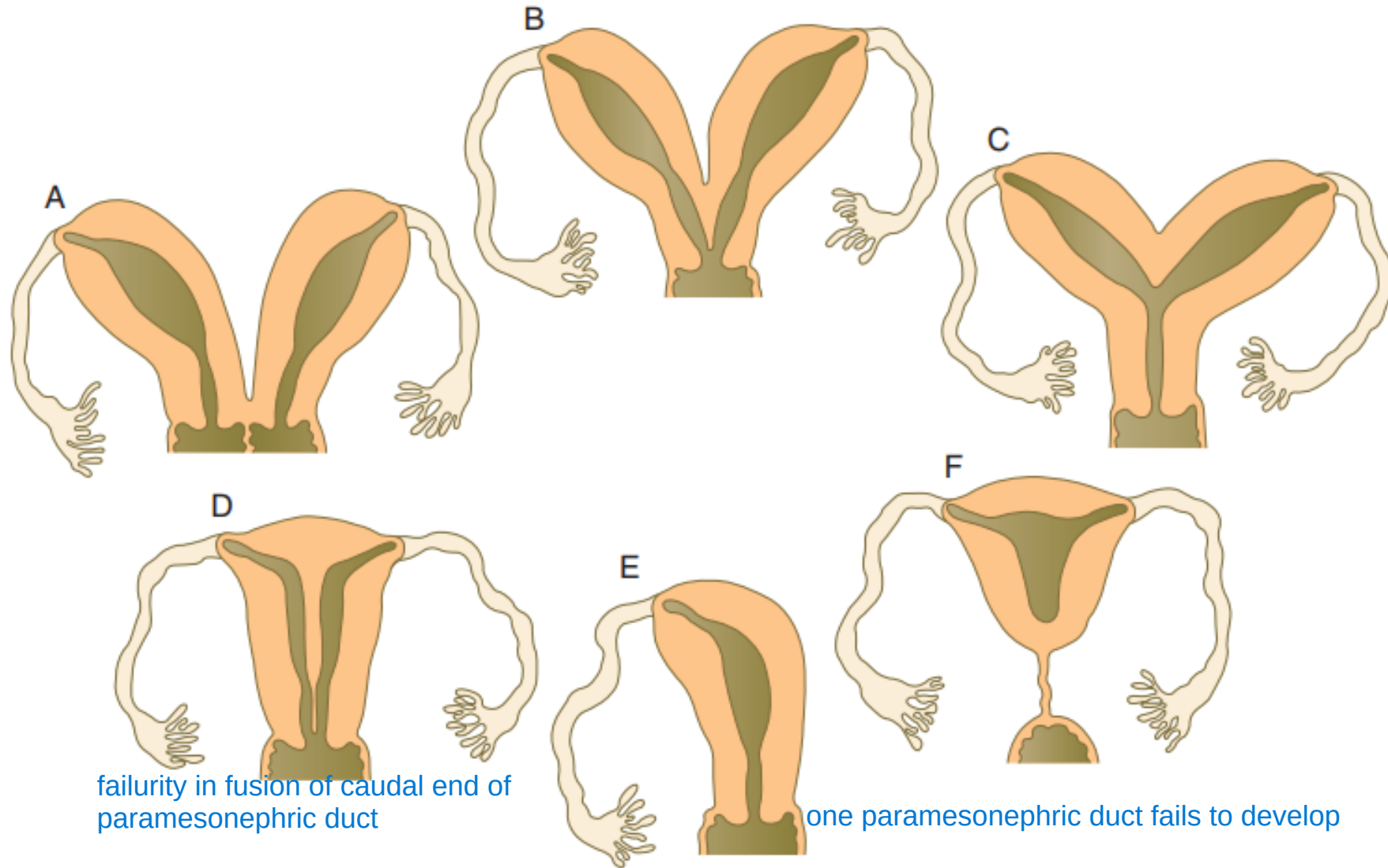


FIGURE 16-37. Abnormalities of the uterus and vagina. **A**, Double uterus and double vagina. **B**, Double uterus and single vagina. **C**, Bicornuate uterus. **D**, Septate uterus. **E**, Unicornuate uterus. **F**, Atresia of the cervix.

Hypospadias

the urethral folds fail to fuse completely

Hypospadias is a birth defect in boys where the opening of the urethra (the tube that carries urine from the bladder to the outside of the body) is not located at the tip of the penis.

epispadias is rare and is usually associated with exstrophy of bladder



Subcoronal



Midshaft



Penoscrotal

Cryptorchidism

- Undescended testes are common in premature males and are seen in about 3% of term males
- Normally, the testes of these individuals descend into the scrotum within the first few months after birth
usually 3 months after birth
- Cryptorchidism results in sterility because spermatogenesis does not normally occur at the temperature of the body cavity
- There is also a 50-fold greater incidence of malignancy in undescended testes

*Thank
you*

